

ROAD TOWARDS SRV6

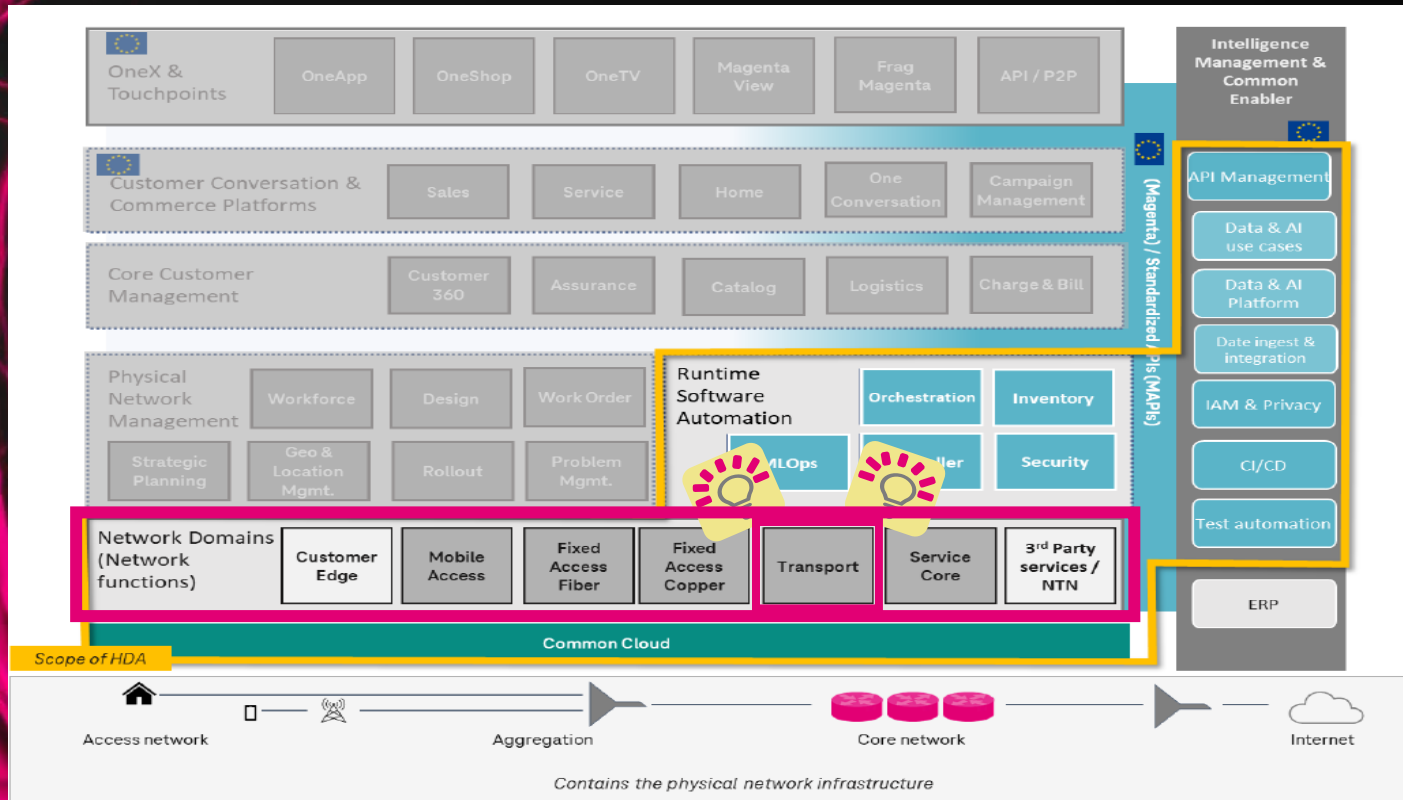
CISCO SRV6 ROADSHOW

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CONTEXT AND MOTIVATION

DT STRATEGY: Horizontal Digital Architecture



Why ?

Single and simplified end-to-end IP transport across domain borders

+

Unified Technology and Provisioning.
Reduction of transport protocols in Core and Data Centers.

+

Future proof strategy and
architecture guideline for DT
footprint.

In-line with DT's overall IPv6 strategy.

WHY NOW?

Requirements for today's networks are changing

Size of networks and scalability – the need to scale up to 50.000 edges, DC integration up to the host, disaggregated aggregation and service creation, ...

Requirements only partially met by MPLS

MPLS proven to complex for end-to-end deployments (Seamless MPLS, MPLS-to-the-Data Center). High scalabilities needs domain-based approach, ...

Goal is a future proof, stable and highly scalable network architecture.

SRv6 as mature technology

Vendors start to prefer SRv6 over MPLS, new features being implemented on SRv6 first (e.g., path tracing). Seamless migration from MPLS, can keep existing services such as L3VPN.

Closer cooperation within DT Group

Need to offer services across different networks, integration of virtualized routers,
Necessary to agree on a common architecture for future networks.

SETUP

- Group wide activity to align on SRv6 across DTs European footprint
 - Addressing, features, standards, RFIs and RFQs, ...
- Architecture blueprint with the intention to develop a harmonized IP transport architecture for all networks
 - Migration mid- to long-term approach, comes with life cycle process of existing network elements. Specifically, existing SRv6 deployments are not forced to migrate to the new architecture.
- Wide variety of different network architectures and topologies
 - historically grown, every network is different, no greenfield
- Looking into a cross-domain concept to be able to (seamlessly) interconnect different service providers
 - Harmonized services across several domains (countries)
 - Also assumes, that we do not need to renumber networks (avoid mistake made with IPv4 and/or RFC1918 addresses)



USE CASES

FlexAlgo Implementation

- 0 – standard SPF
- 1 – strict SPF as being used in traffic-engineering and avoidance of loops
- 2 – latency optimization
- 3 and 4 – disjoint paths in the A/B-plane topology

End-to-End SRv6/DC Integration

- Extension of backbone into the datacenter (e.g., EVPN)
- SRv6 up to the hosts
- Replacement of VXLAN

SRv6 in 6G

- Simplification of existing GTP stack in 3GPP
- Pushed into 3GPP for better convergence/integration into SRv6 network design

SRv6 as Enabler/Basis for new Network Designs

- Network extensions beyond physical hardware, integration of virtualized routers in Cloud environments via SRv6 connectivity

SRv6 as technology of choice for extension

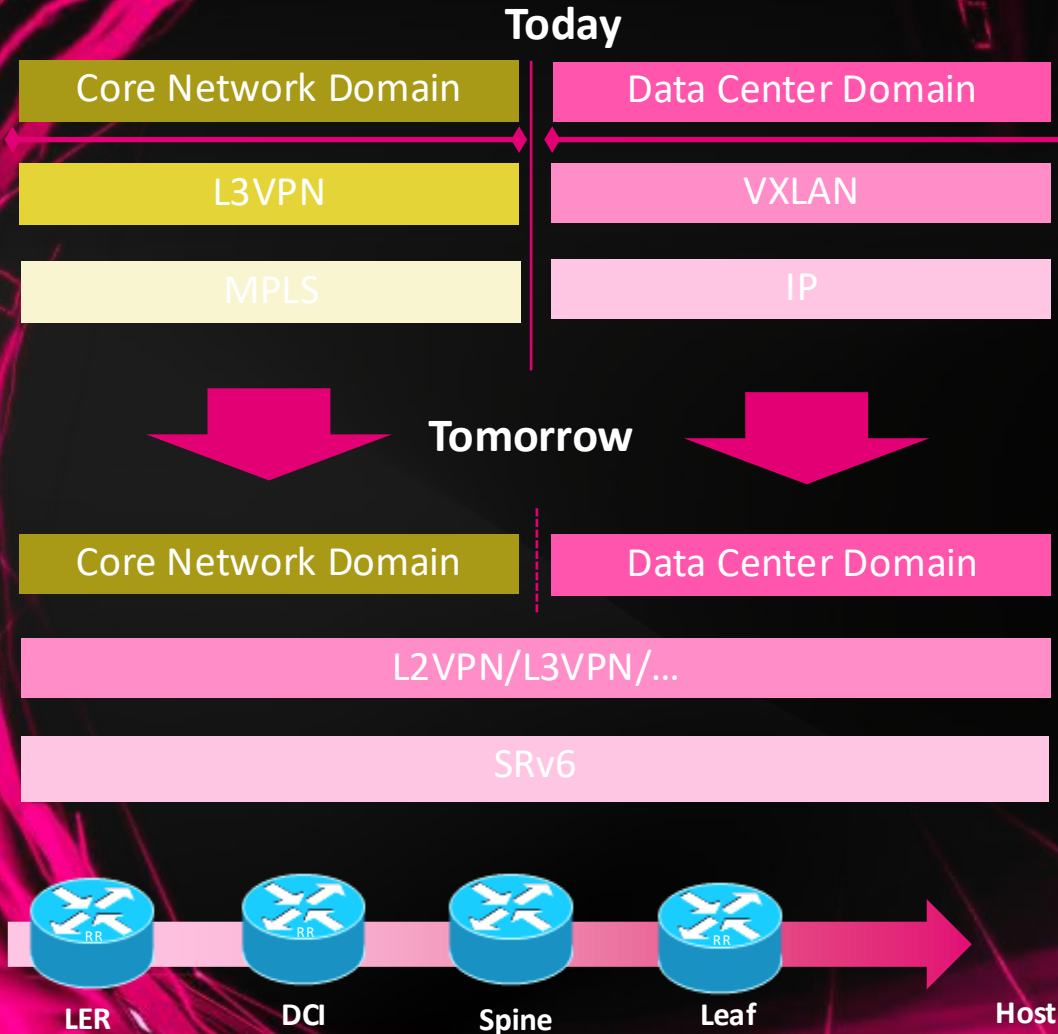
Multivendor/Standards are Key

- Starting point are about 10 Core Networks with different architectures and vendors
- Open Standards (e.g., IETF) and interoperability are mandatory
- Goal: harmonized feature set across all countries/NatCos (at least 99%)

PoCs across different NatCos/Countries

- NGMA - Next Generation Mobile Access
- SDBH – Software Defined Business HUB
- PLE with SRv6 –Private Line Emulation
- Deployment in Poland

DATACENTER INTEGRATION

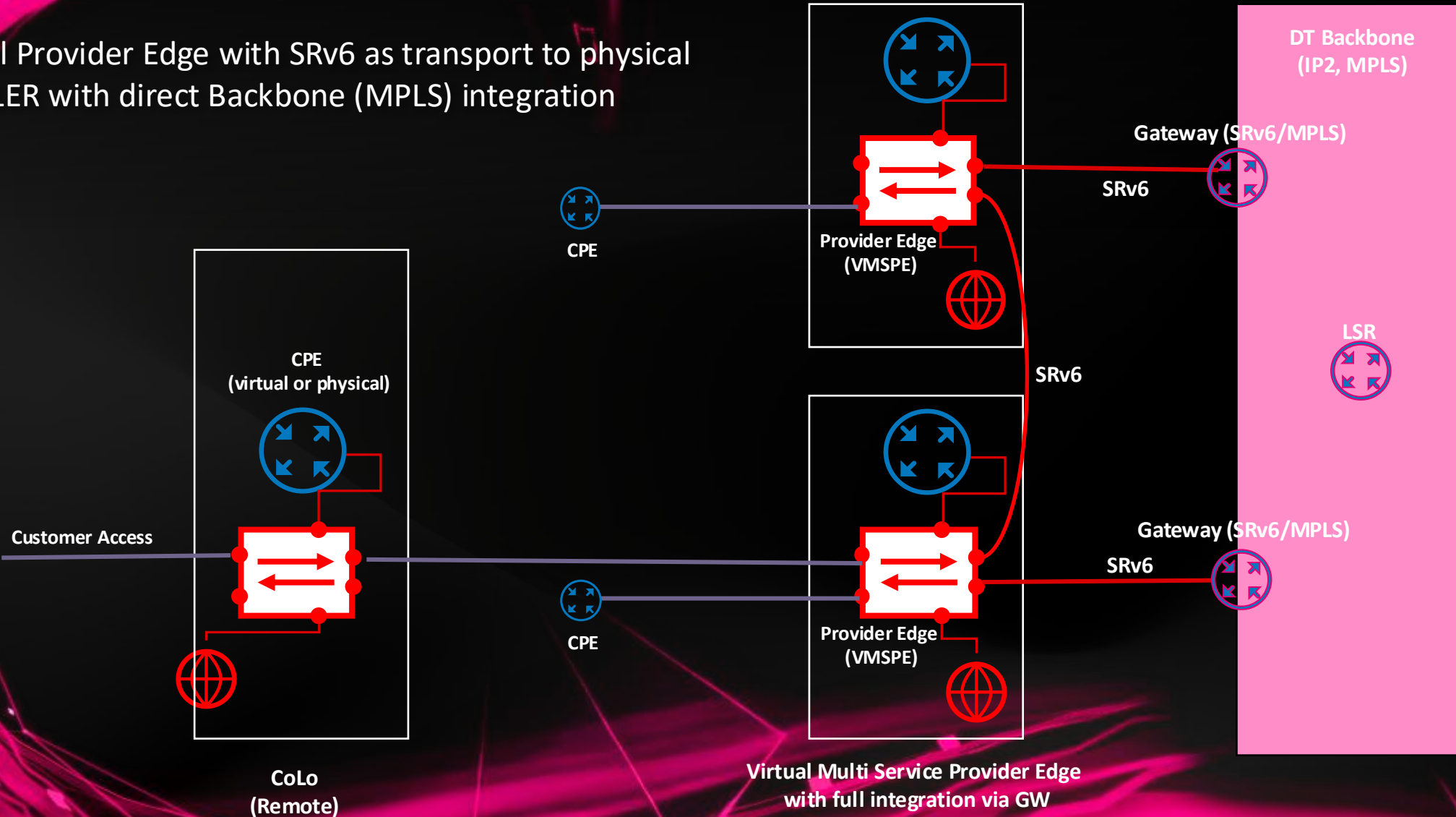


Datacenter Integration with SRv6

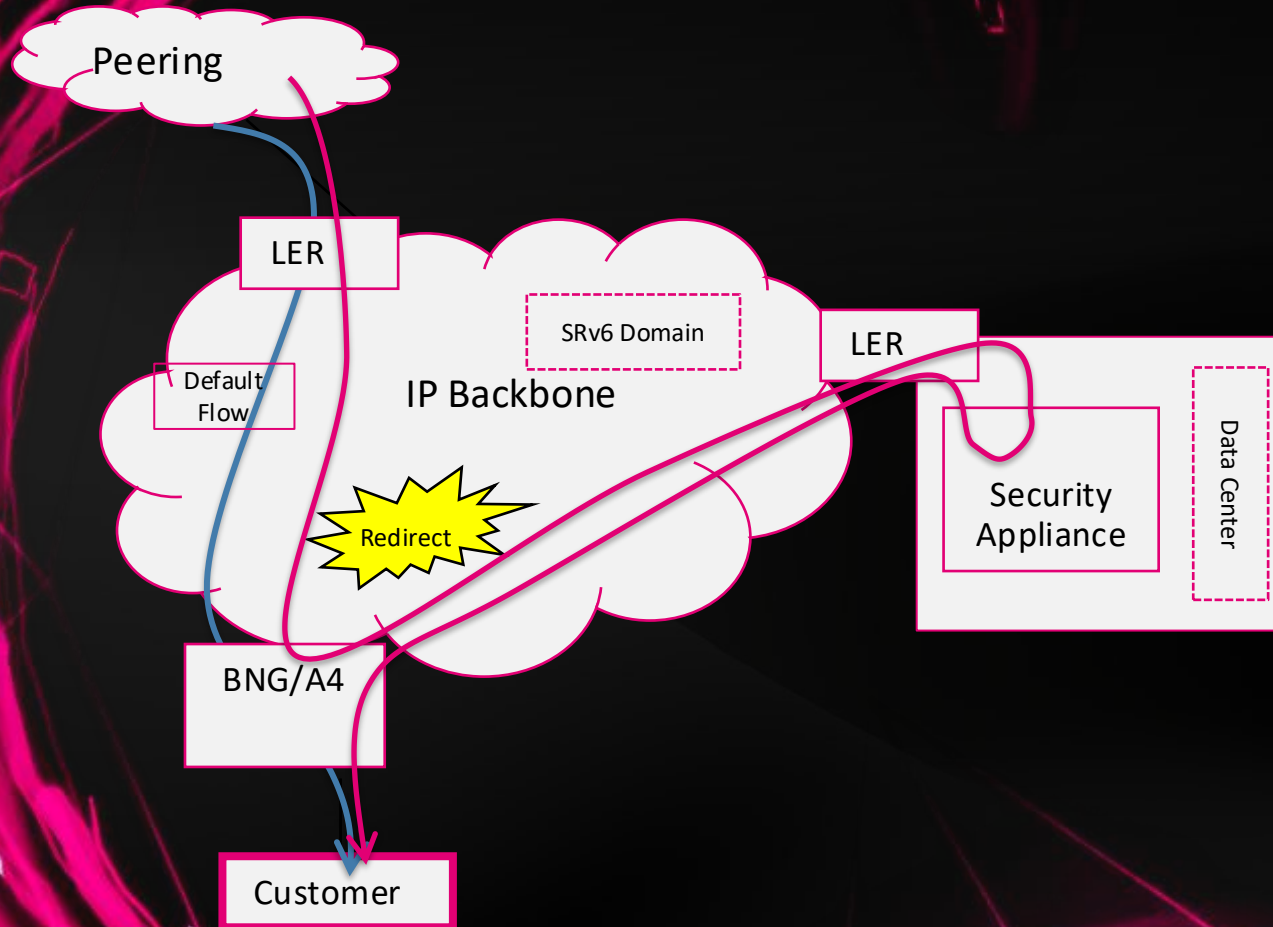
- We need to define “Datacenter Integration”
 - Different understandings of “Integration”
 - Underlay, Overlay/Service, Both?
 - Integration of Underlay/Overlay/Services?
 - Simplified deployment
 - SRv6 up to the host, how far do we go with SRv6?
 - Core and DC use same technology, high flexibility in positioning the “edge”
 - Same technologies for all domains + convergence towards single protocol (SRv6)
 - Role of DCI, Spine & “LER” in End-to-End Scenarios can be re-evaluated, no need to daisy-chain dozens of boxes
- ...

USE CASE: NETWORK EXTENSION

Virtual Provider Edge with SRv6 as transport to physical LER with direct Backbone (MPLS) integration



USE CASE: INLINE SECURITY



Seamless Integration of Security Appliance

- Activation of traffic redirection in case of DDoS attack
- Traffic routed via physical/virtual security appliance to clean up traffic
- Optionally service chaining inside DC (also based on SRv6)

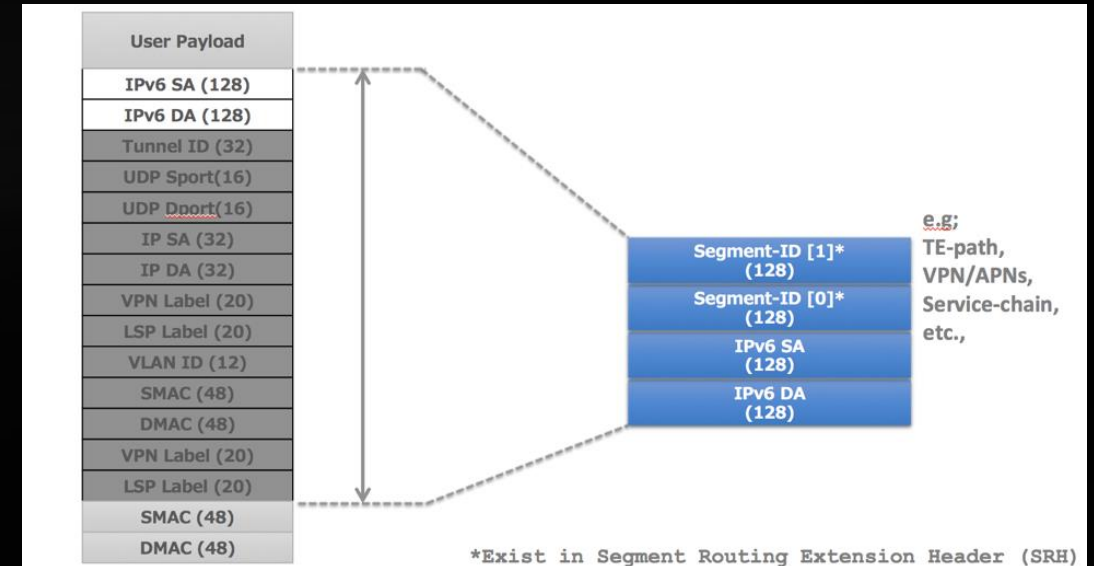
Benefit & Status

- One of the standard use cases for FlexAlgo, easy to extend
 - Can be implemented “on top” of the existing infrastructure
 - Standard SRv6 feature for re-routing, optionally seamlessly integrates with SRv6 aware datacenter

USE CASE: SRV6 FOR GTP TUNNELS

Simplification of GTP Tunnels via SRv6

- Replacement of GTP-U with SRv6
 - Simplification of User Plane via native SRv6 encapsulation
 - Reduction of tunnel overhead
 - Can be seen as “Rip-n-replace of GTP with SRv6”
 - Allows seamless integration with network slicing
- Mainly driven in 3GPP by “classical” router vendors
 - Cisco, Softbank, DT, ...
 - Not much support by mobile vendors
- Softbank has already deployed SRv6 live for their Mobile Network in Japan



- Under discussion in 3GPP for 6G, limited support by mobile vendors
 - Significant change in for existing deployments/protocol stacks
- Seamless integration with End-to-End SRv6 & Service Chaining possible

SUMMARY

- SRv6 as key technology for evolution of all networks across DT group
- Future proof, extremely flexible and feature rich – simplifying ~~today's network architectures~~
- Common forwarding and control plane across network domains (including DC, Access/Aggregation and Backbone, Mobile/Fixed Backhaul).
- Simplification due to removal of network protocols (LDP, NSH, ...).
- SRv6 base standards are available as RFCs. Work on extensions like Header Compression and uSID has a lot of momentum.
- Large number of interoperable implementations including Open Source.
- Evolution from existing MPLS based networks. Ideal migration concept as mixture of SR- and non-SR-capable routers possible.
- SRv6 as driver for 5G networks (e.g., Network Slicing).
- Several large deployments.

Harmonized Strategy

Convergence on a single forwarding paradigm.

Well defined standards and interoperable implementations.

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